THE HYPERBOLA

CONICAL SECTIONS

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Hyperbola with Center at the Origin

Transverse axis is on the x-axis

Transverse axis is on the y-axis

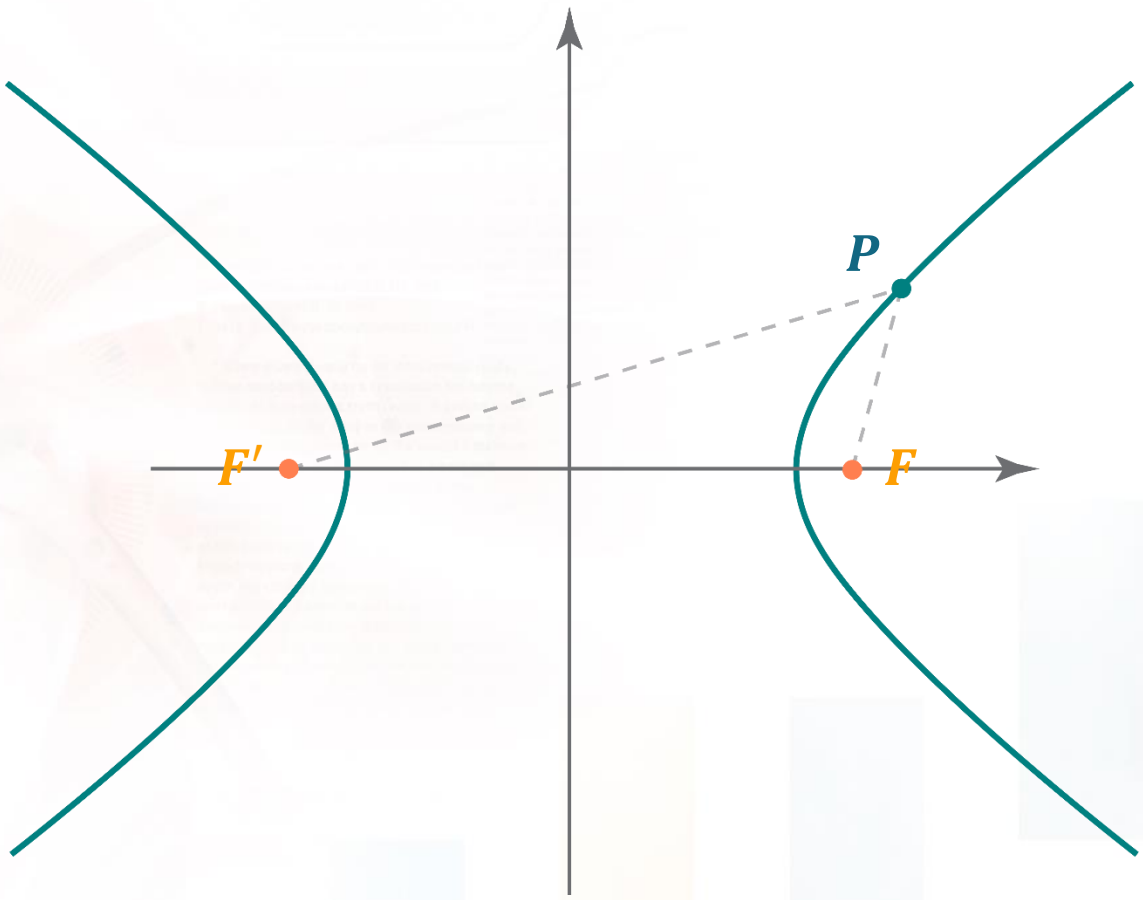
Hyperbola with Center at $C(h, k)$

Transverse axis is along the x-axis

Transverse axis is along the y-axis

HYPERBOLA WITH CENTER AT THE ORIGIN

A **hyperbola** is the set of all points in a plane, the **difference** of whose distances from two fixed points (the **foci**) in the plane is a positive **constant** ($2a$).



$$d(P, F') - d(P, F) = 2a$$

CENTER AT THE ORIGIN

center,

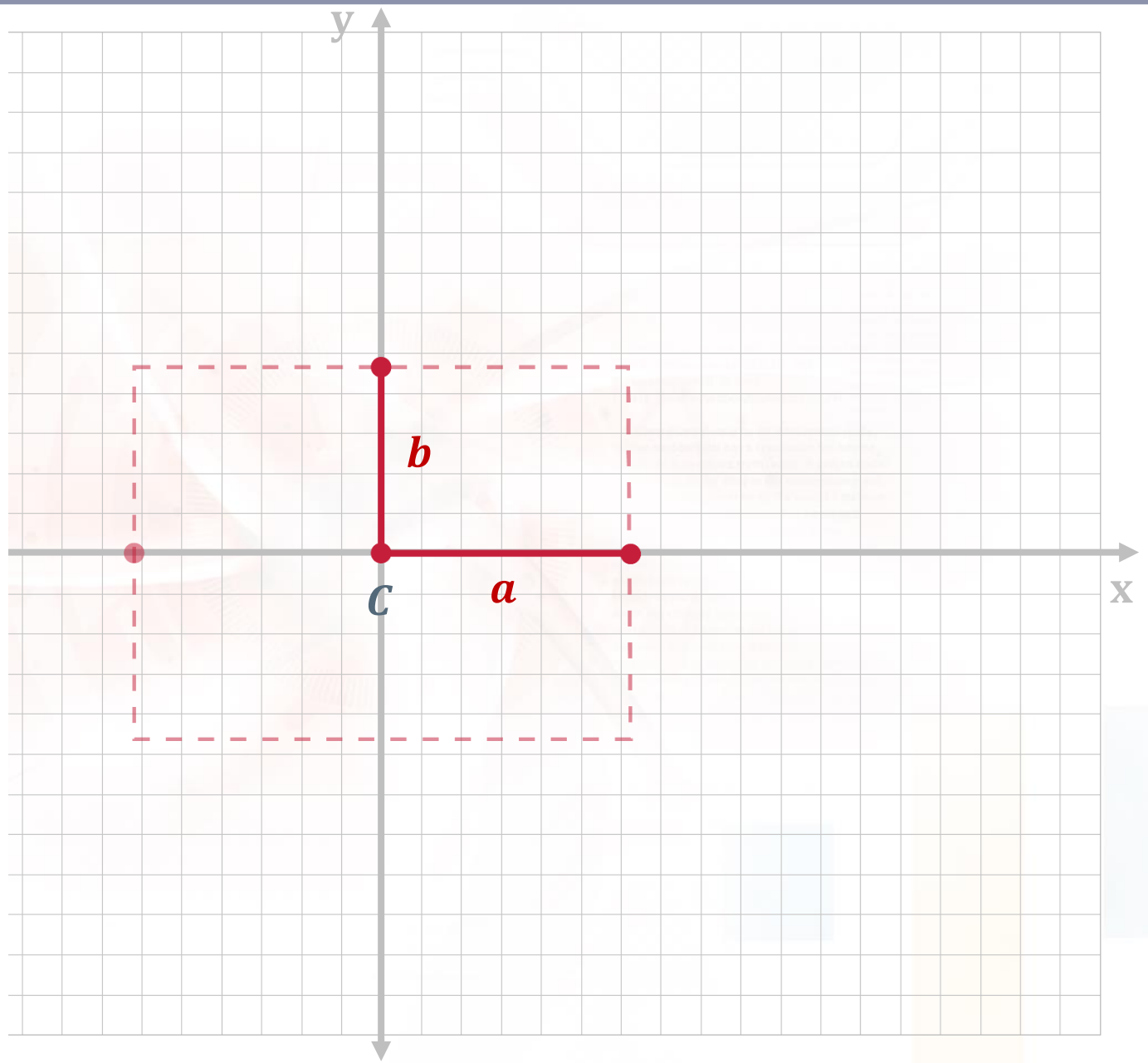
$C(0, 0)$

transverse axis,

$2a$

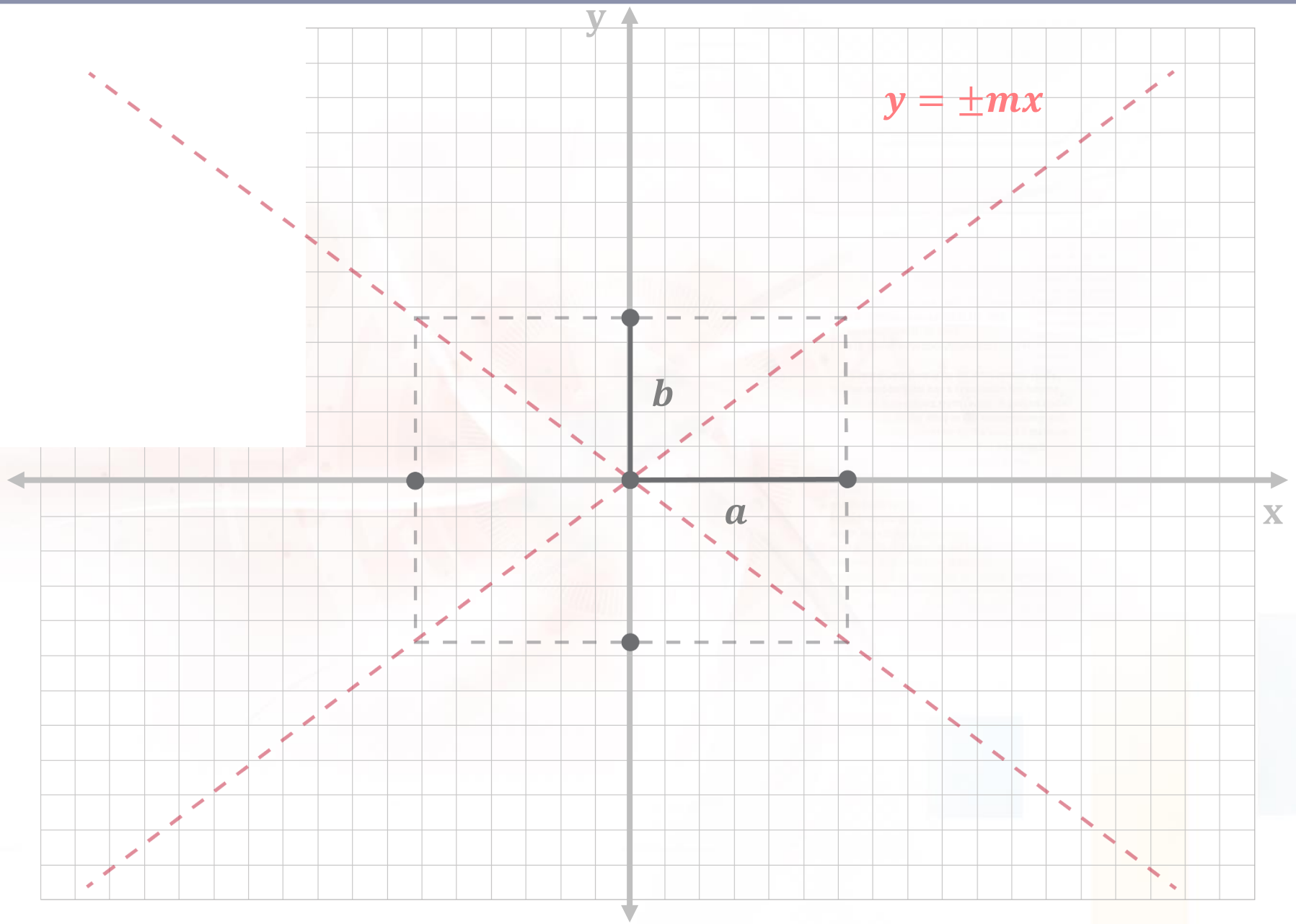
conjugate axis,

$2b$



asymptotes,

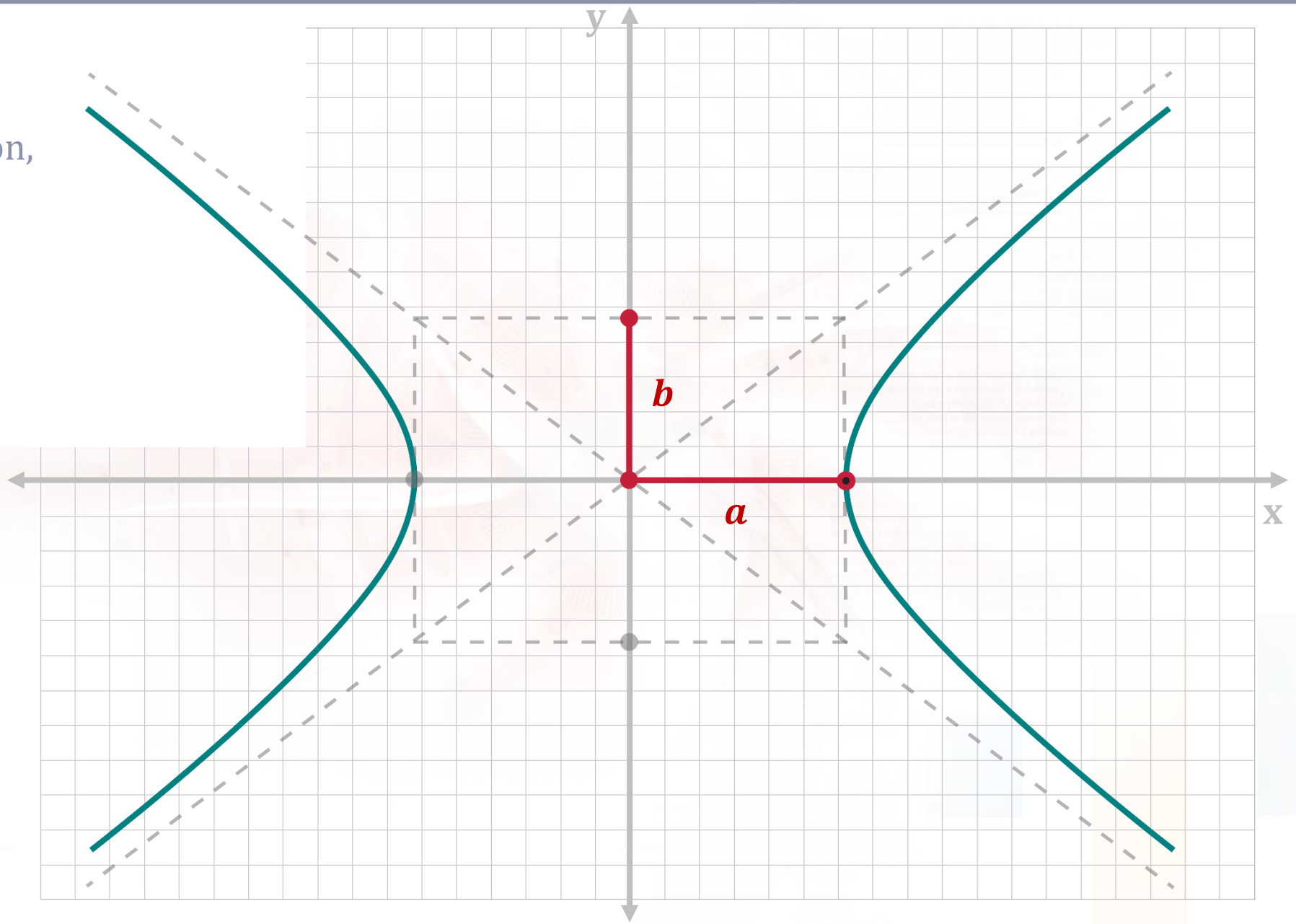
$$y = \pm \frac{b}{a} x$$



STANDARD EQUATION

Standard Equation,

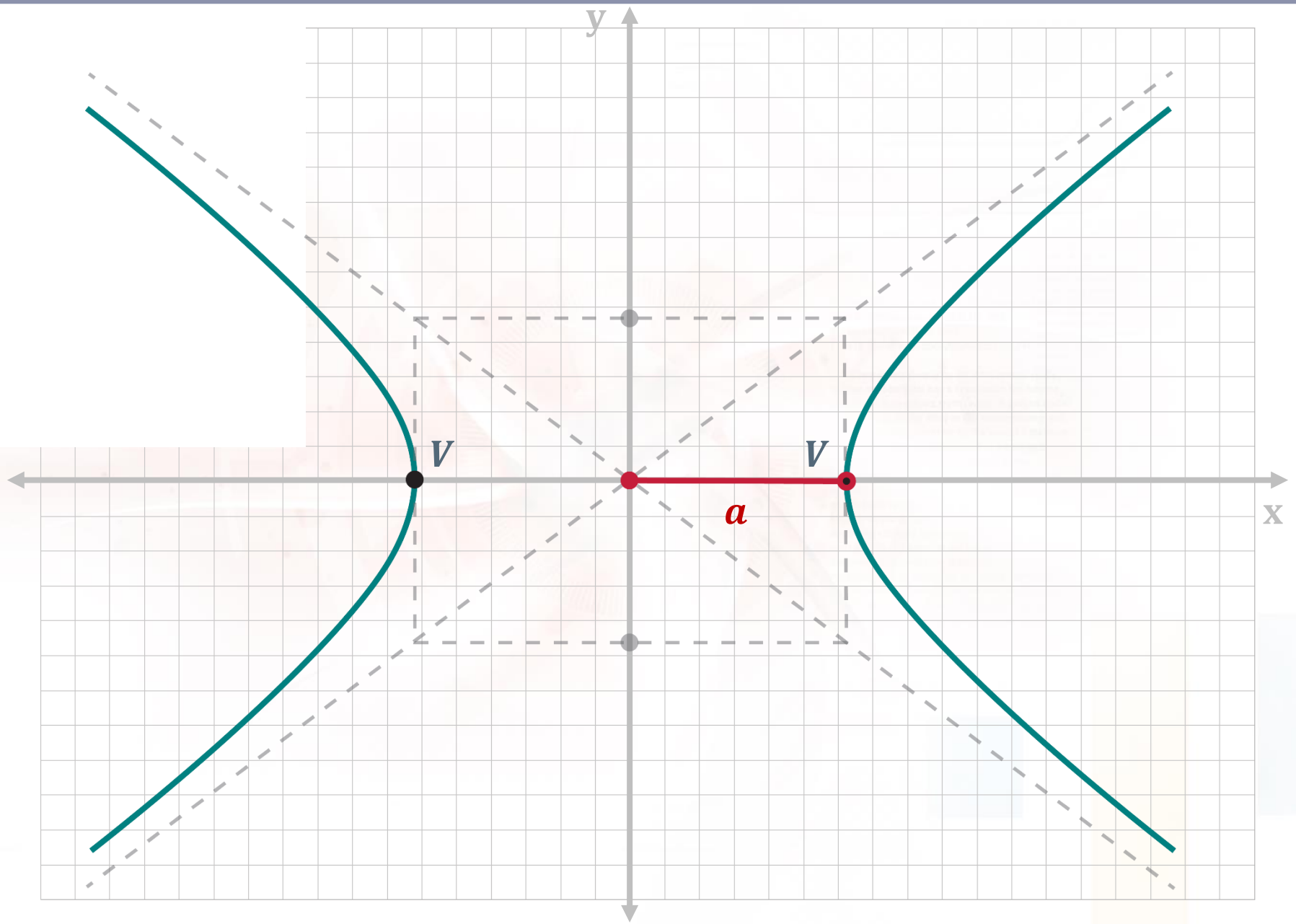
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



VERTICES

vertices,

$V(\pm a, 0)$

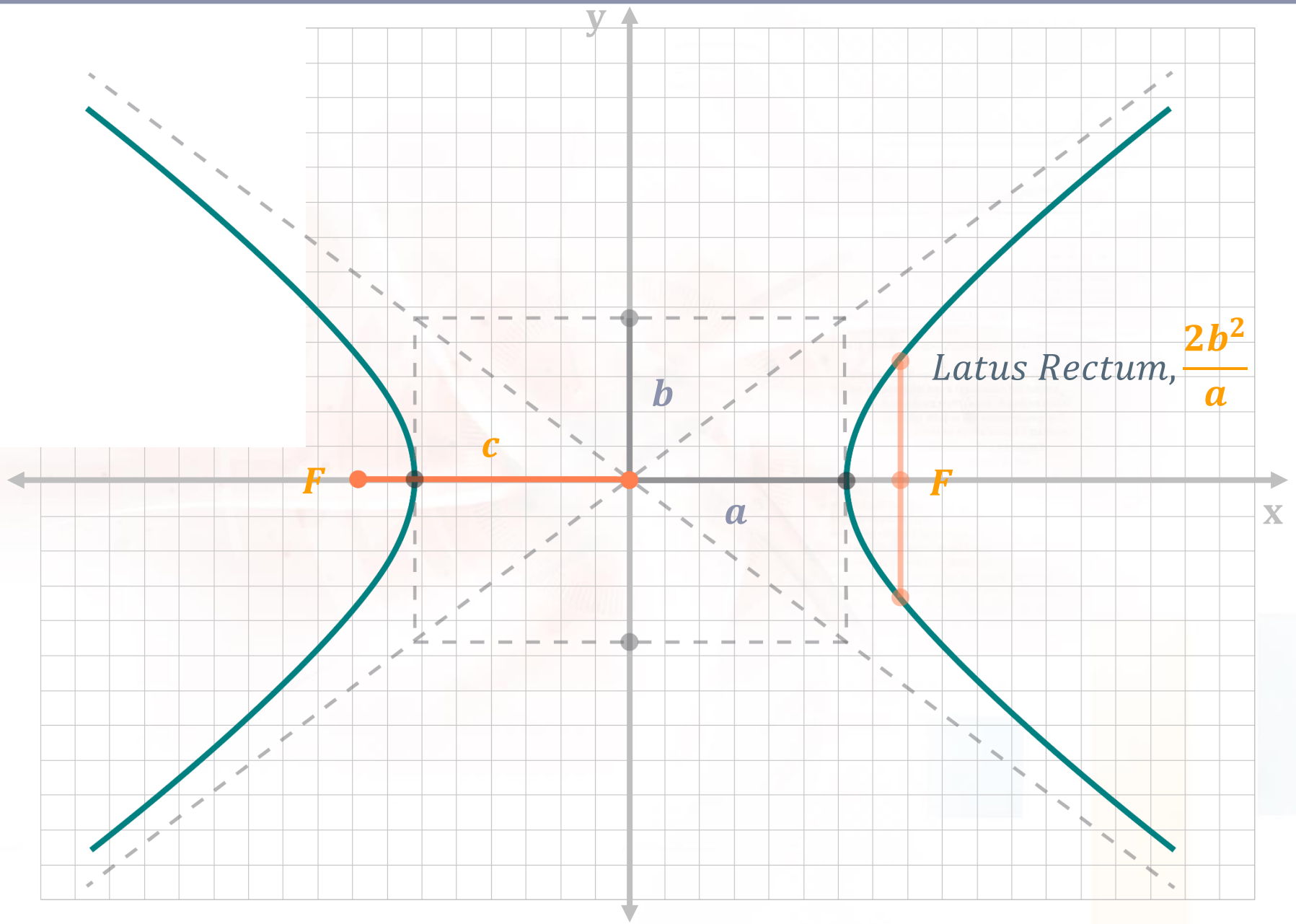


FOCI

foci,

$F(\pm c, 0)$

$c^2 = a^2 + b^2$



TRANSVERSE AXIS is on the $x - axis$

Standard Equation,

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(0, 0)$

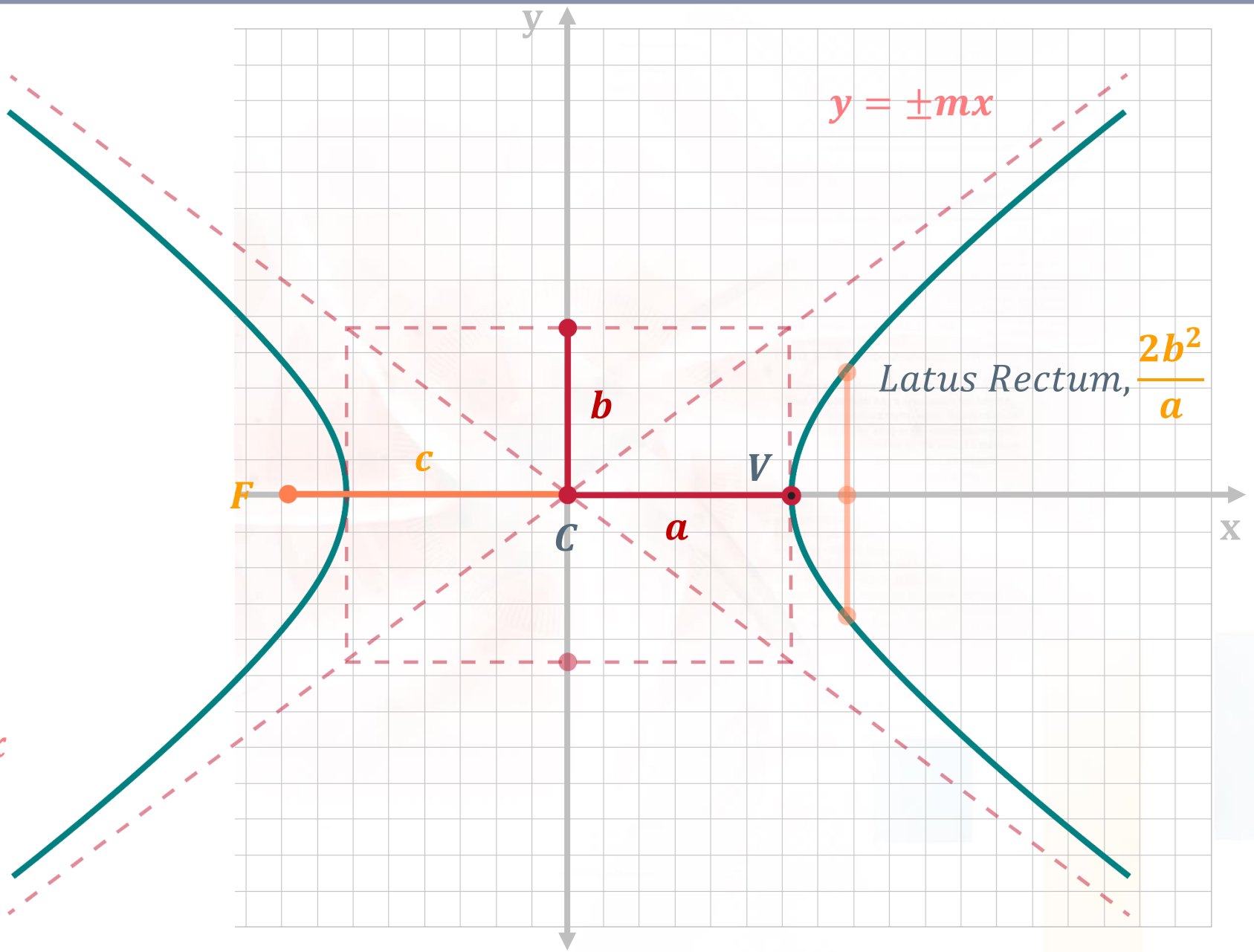
vertices, $V(\pm a, 0)$

foci, $F(\pm c, 0)$

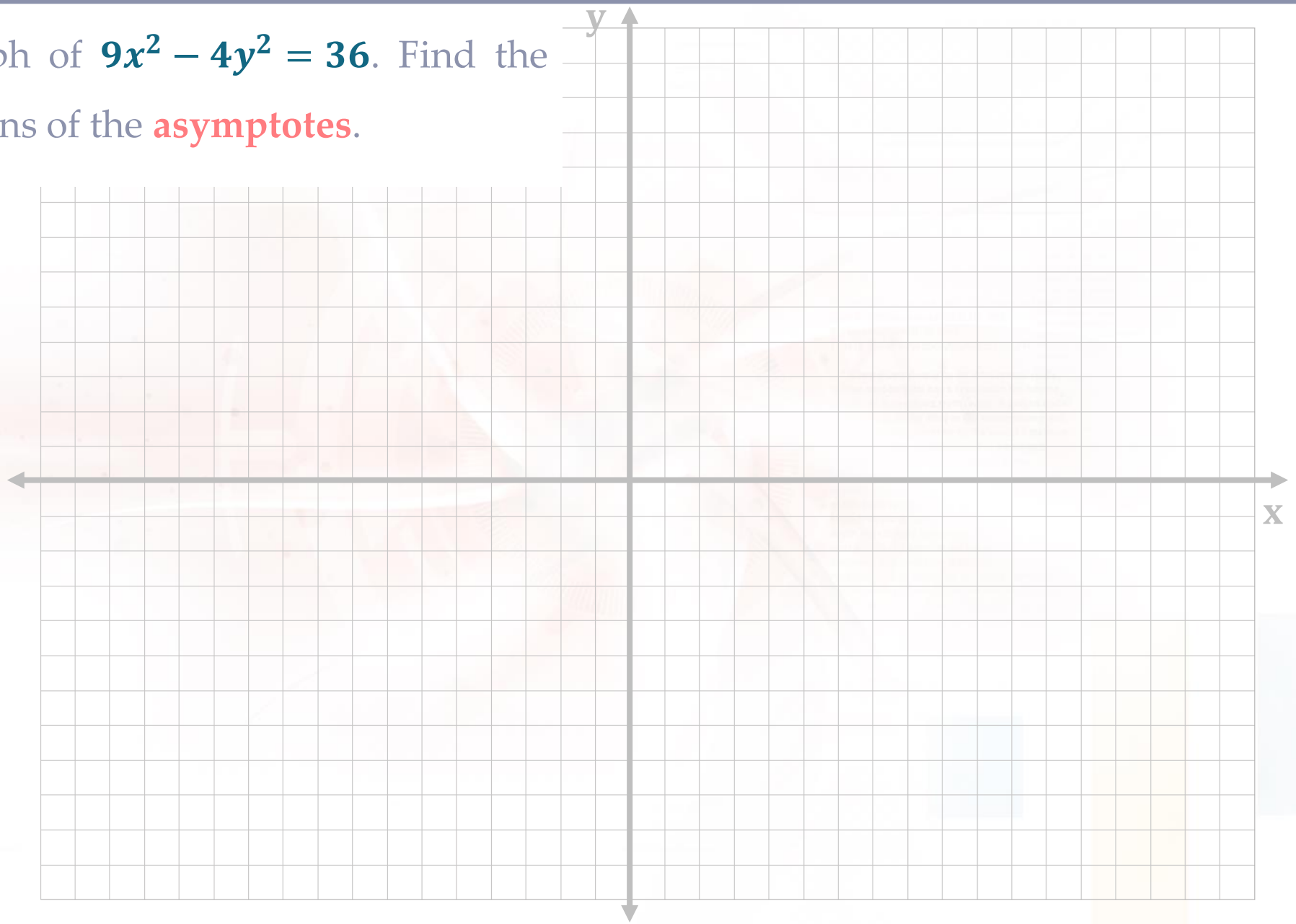
asymptotes, $y = \pm \frac{b}{a} x$

transverse axis, $2a$

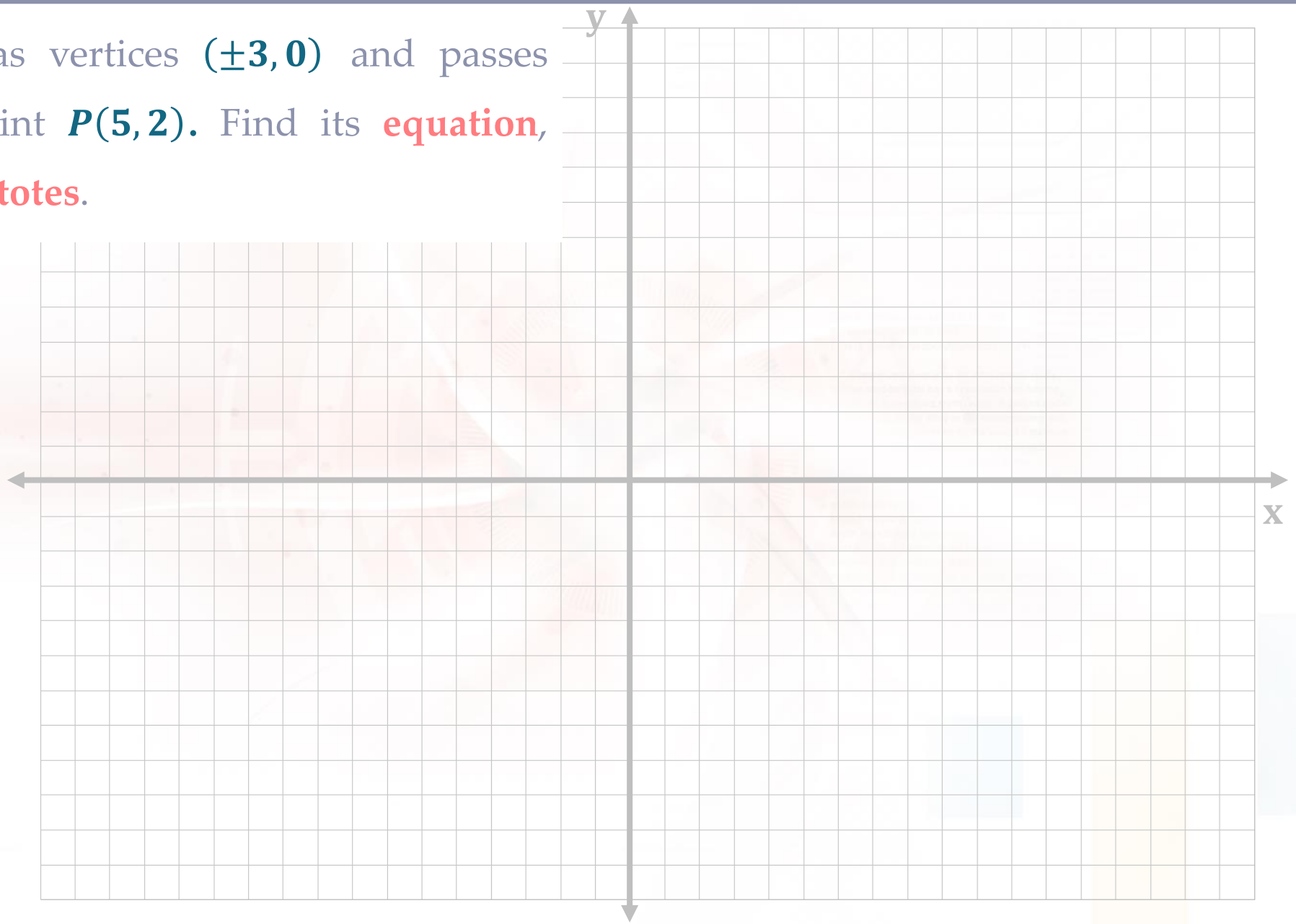
conjugate axis, $2b$



Sketch the graph of $9x^2 - 4y^2 = 36$. Find the **foci** and equations of the **asymptotes**.



A hyperbola has vertices $(\pm 3, 0)$ and passes through the point $P(5, 2)$. Find its **equation**, **foci**, and **asymptotes**.



TRANSVERSE AXIS is on the $y - axis$

Standard Equation,

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(0, 0)$

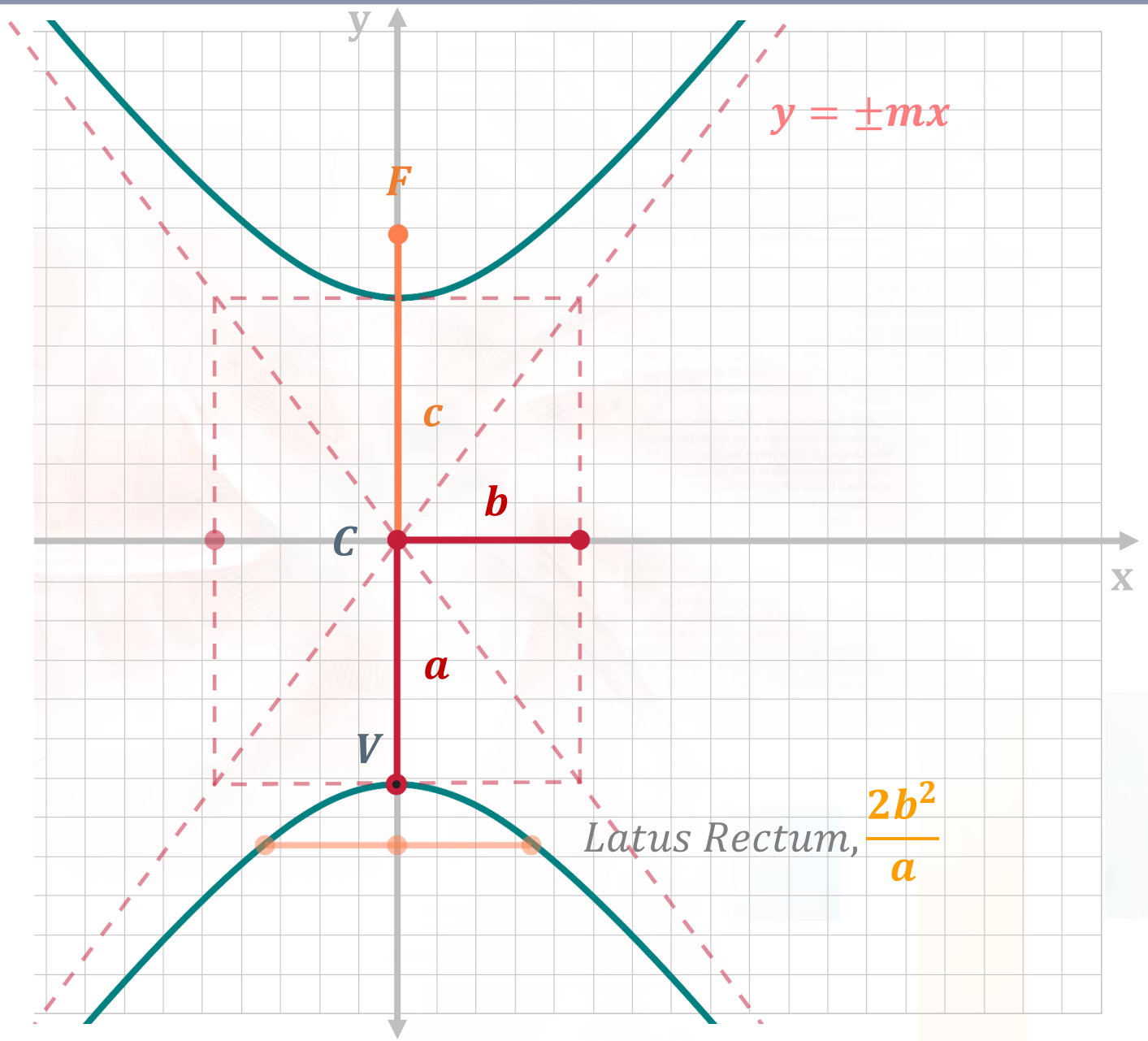
vertices, $V(0, \pm a)$

foci, $F(0, \pm c)$

asymptotes, $y = \pm \frac{a}{b} x$

transverse axis, $2a$

conjugate axis, $2b$



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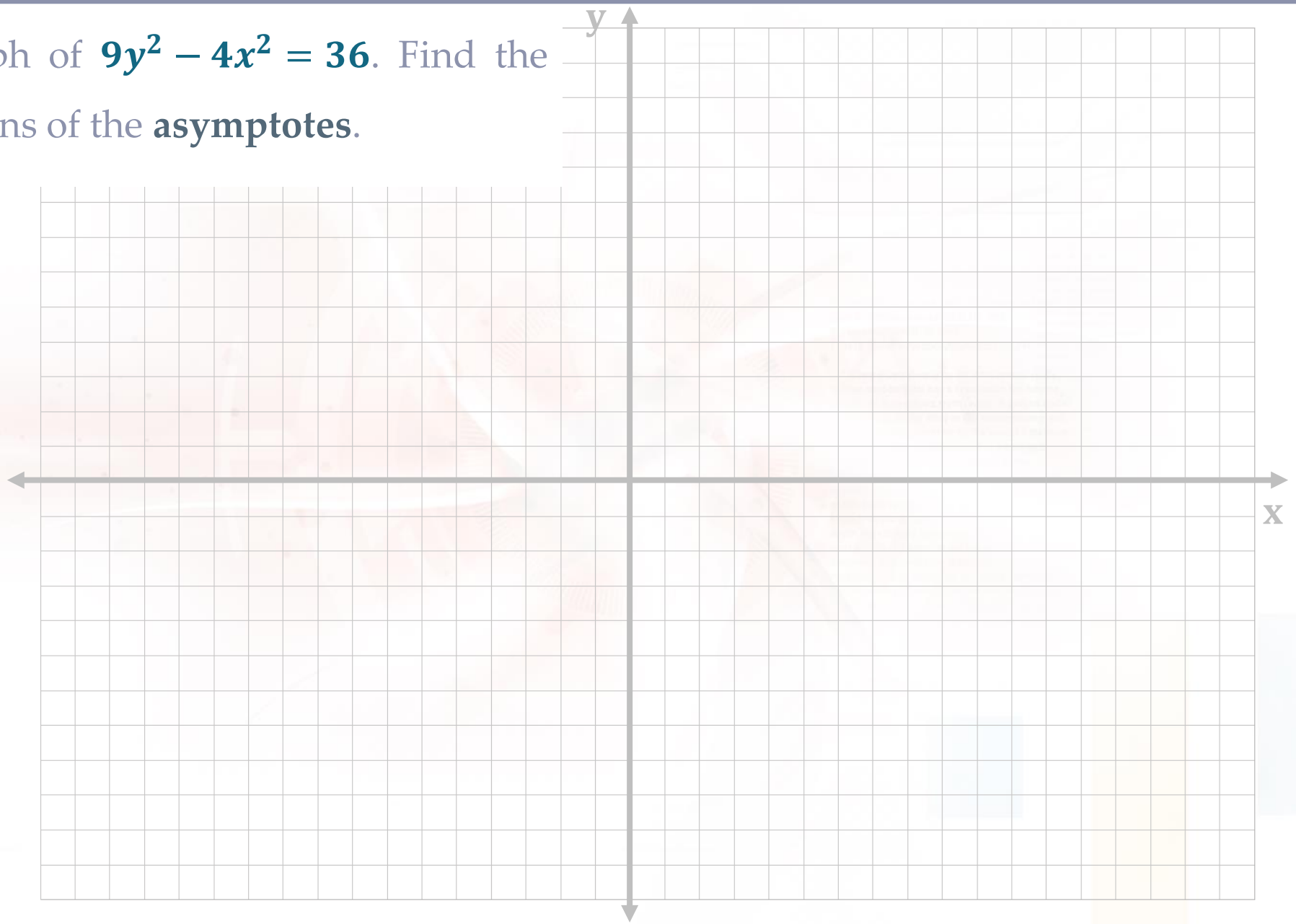
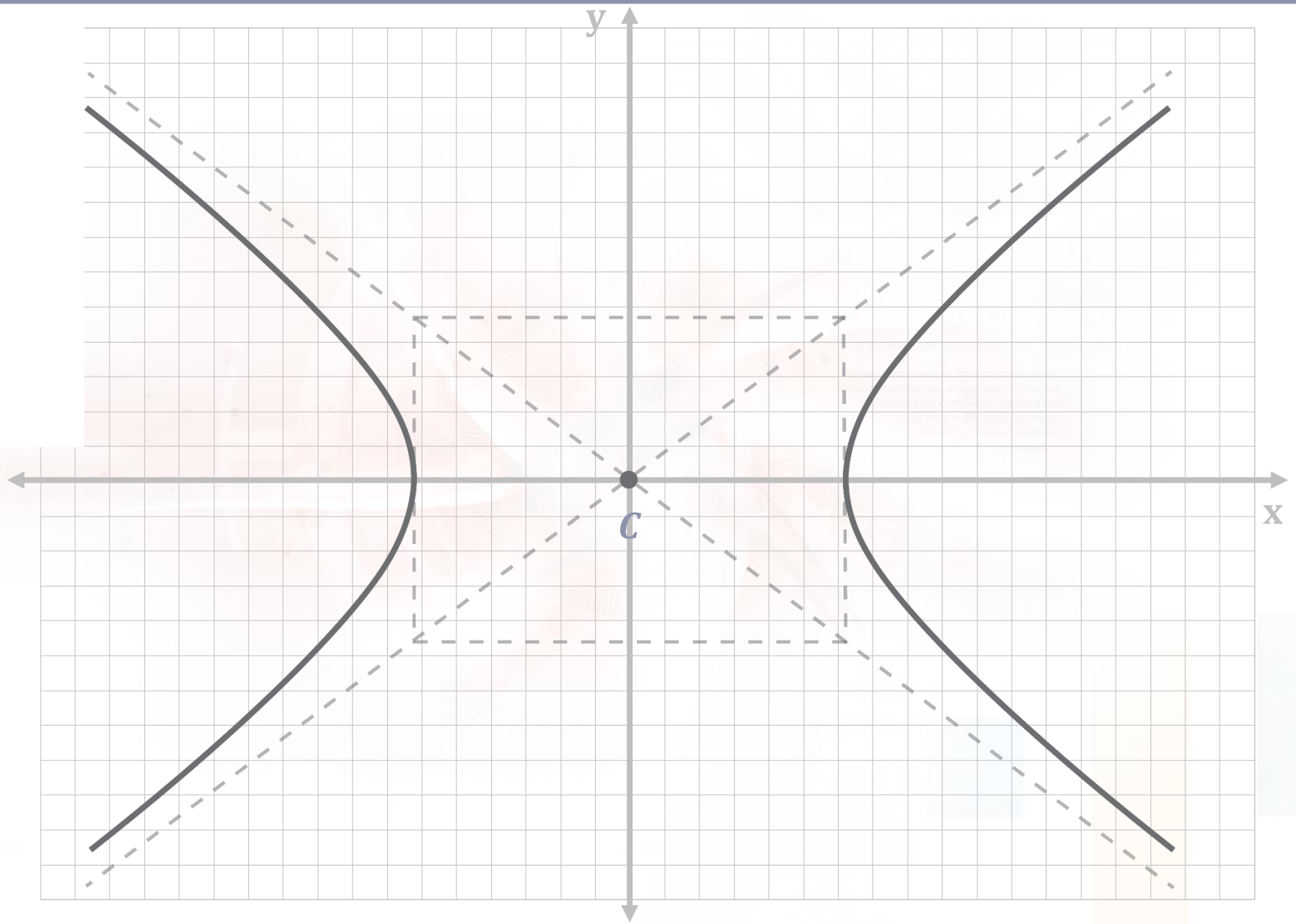


HYPERBOLA WITH CENTER AT $C(h, k)$

CENTER AT THE ORIGIN

center,

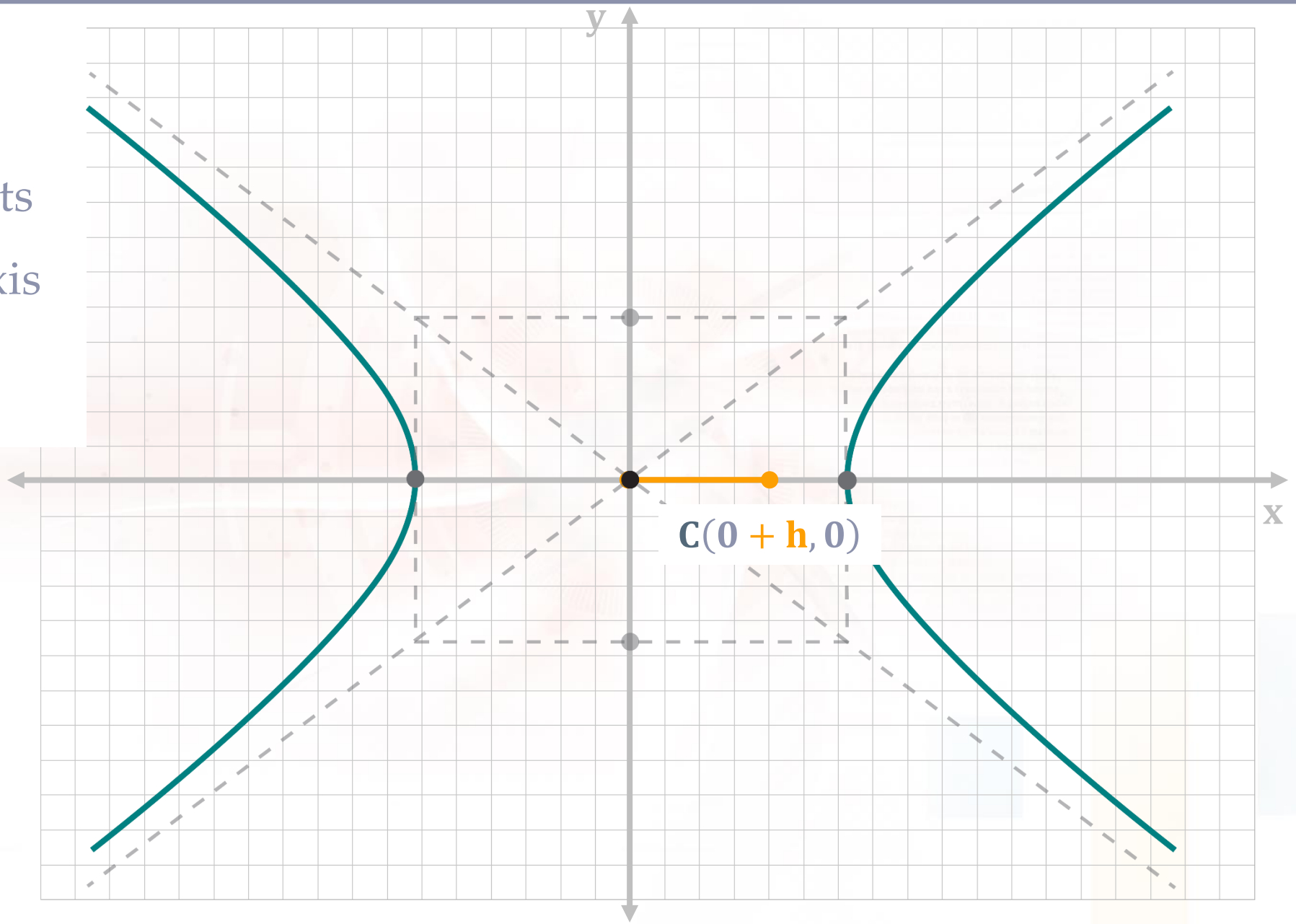
$C(0, 0)$



CENTER AT $C(h, 0)$

moving h units

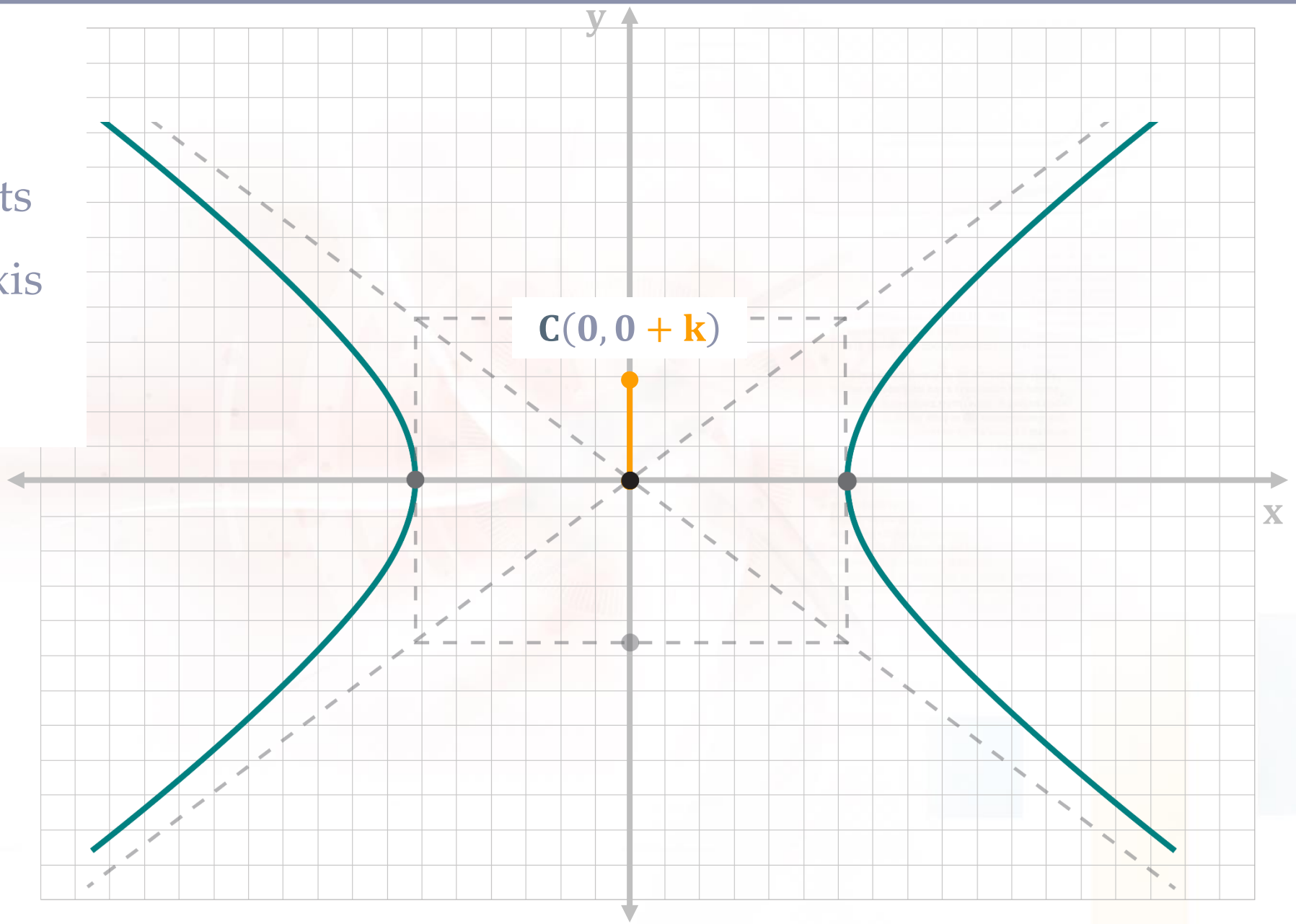
along the x-axis



CENTER AT $C(0, k)$

moving k units

along the y-axis

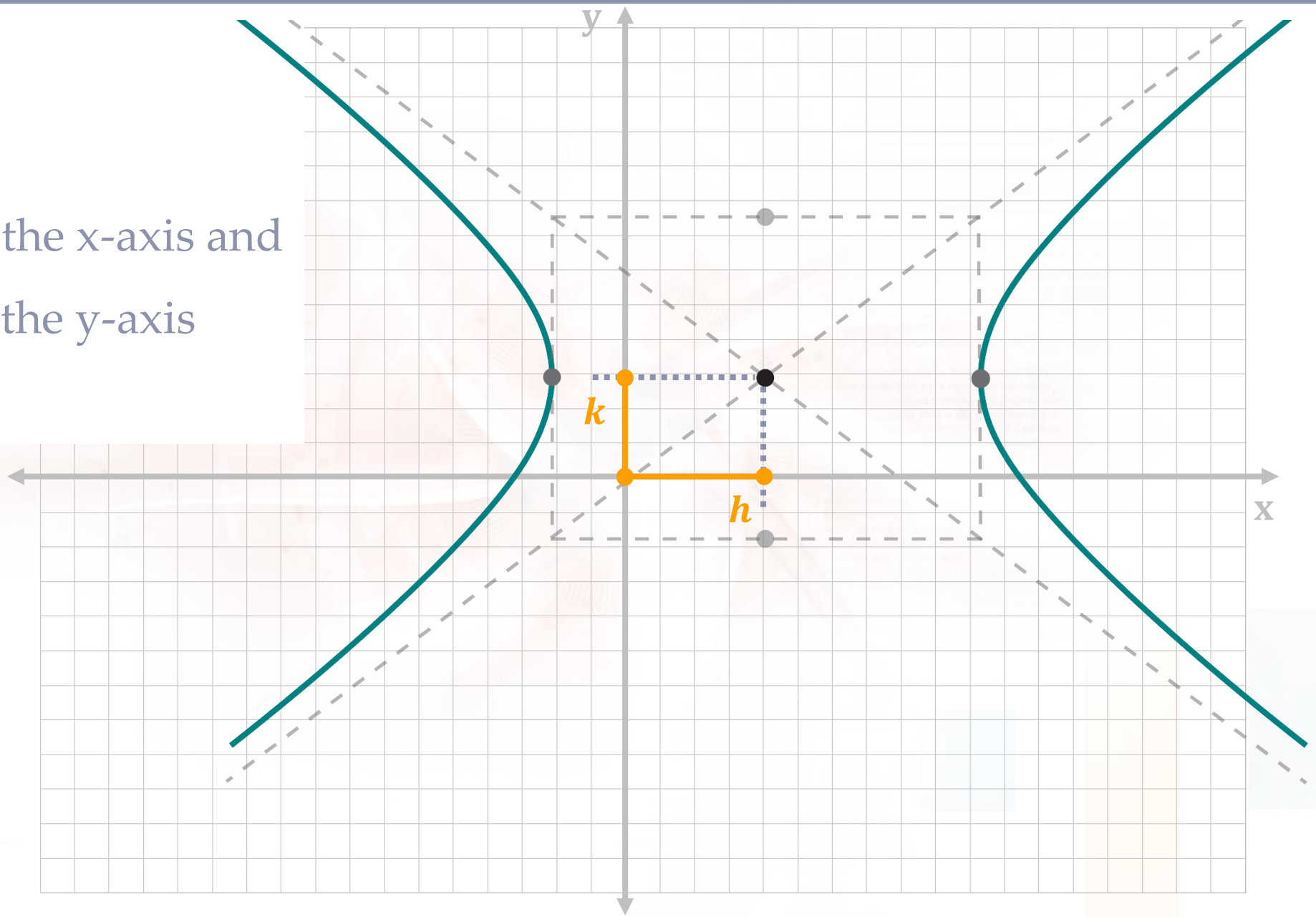


CENTER AT $C(h, k)$

moving

h units along the x-axis and

k units along the y-axis



asymptotes,

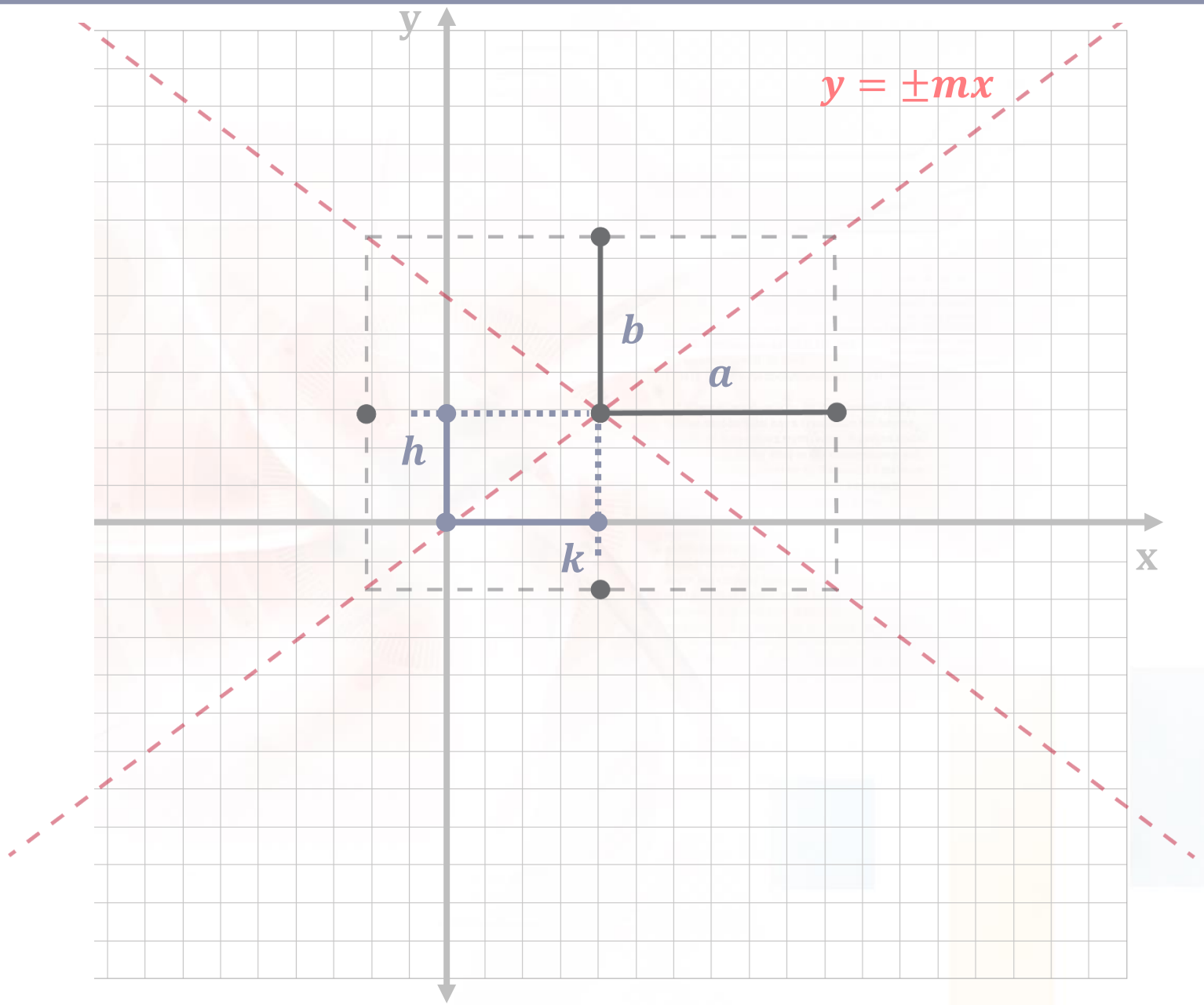
$$(y-k) = \pm\frac{b}{a}(x-h)$$

transverse axis,

$\mathbf{2a}$

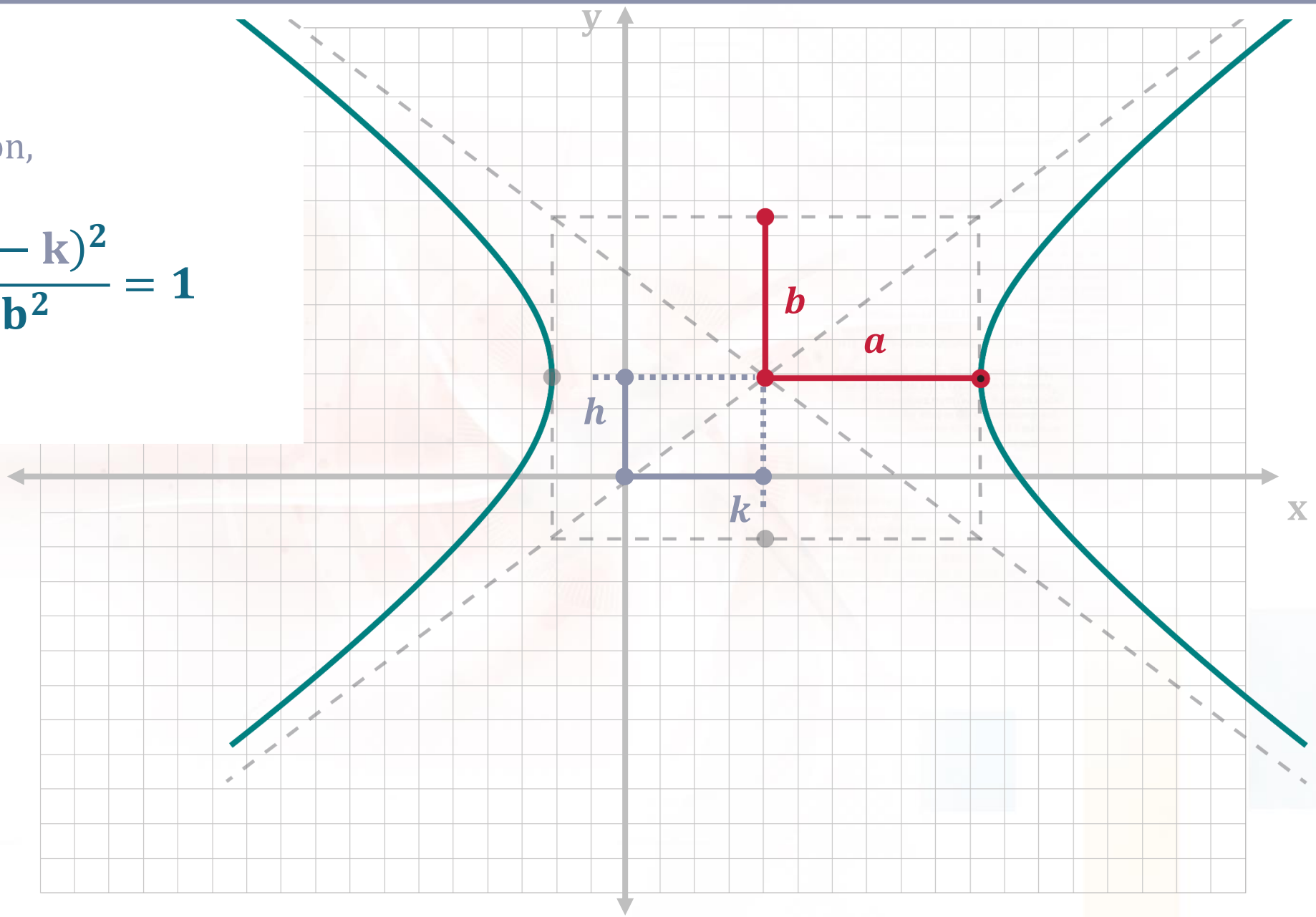
conjugate axis,

$\mathbf{2b}$



Standard Equation,

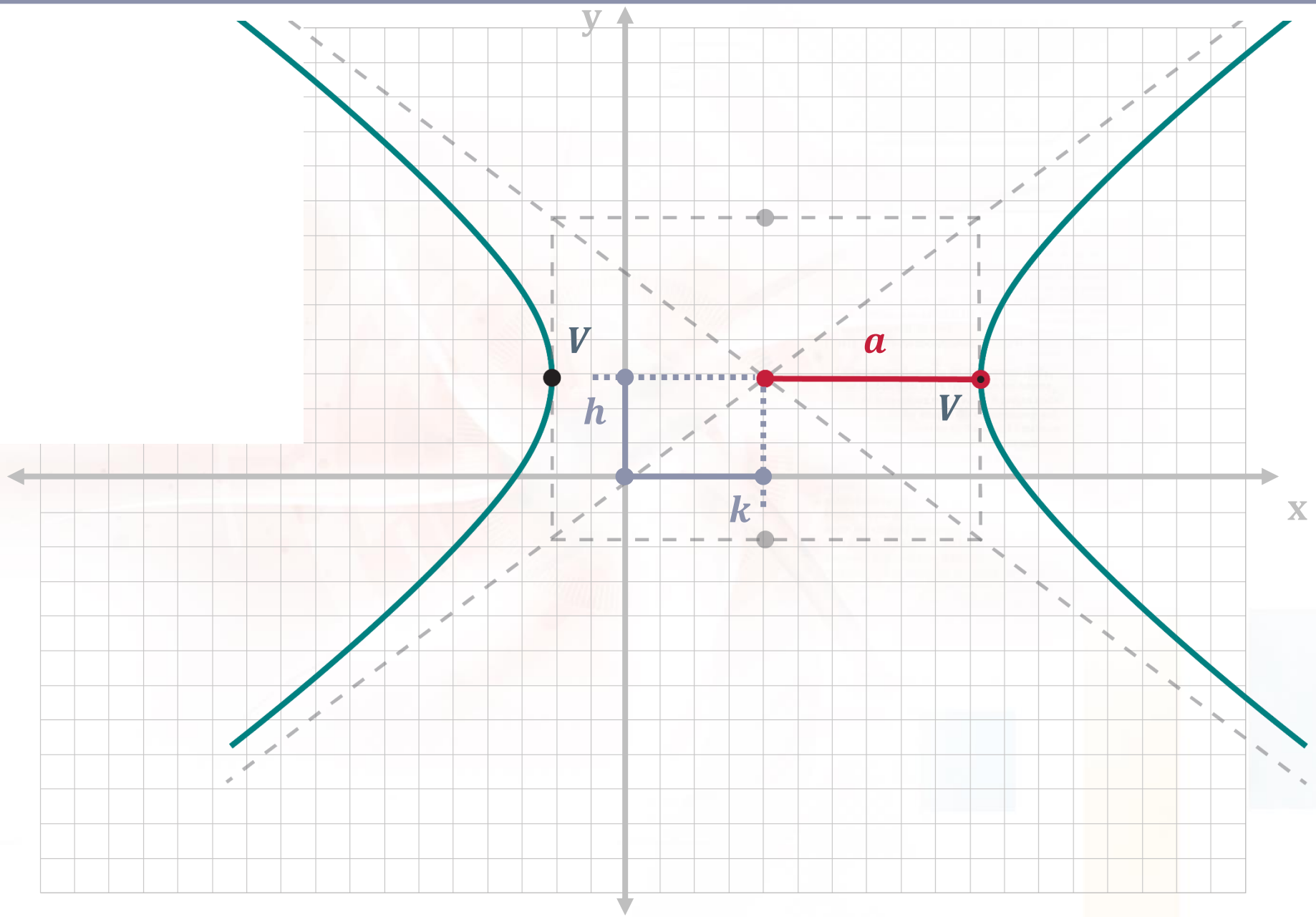
$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$



VERTICES

vertices,

$$V(h \pm a, k)$$

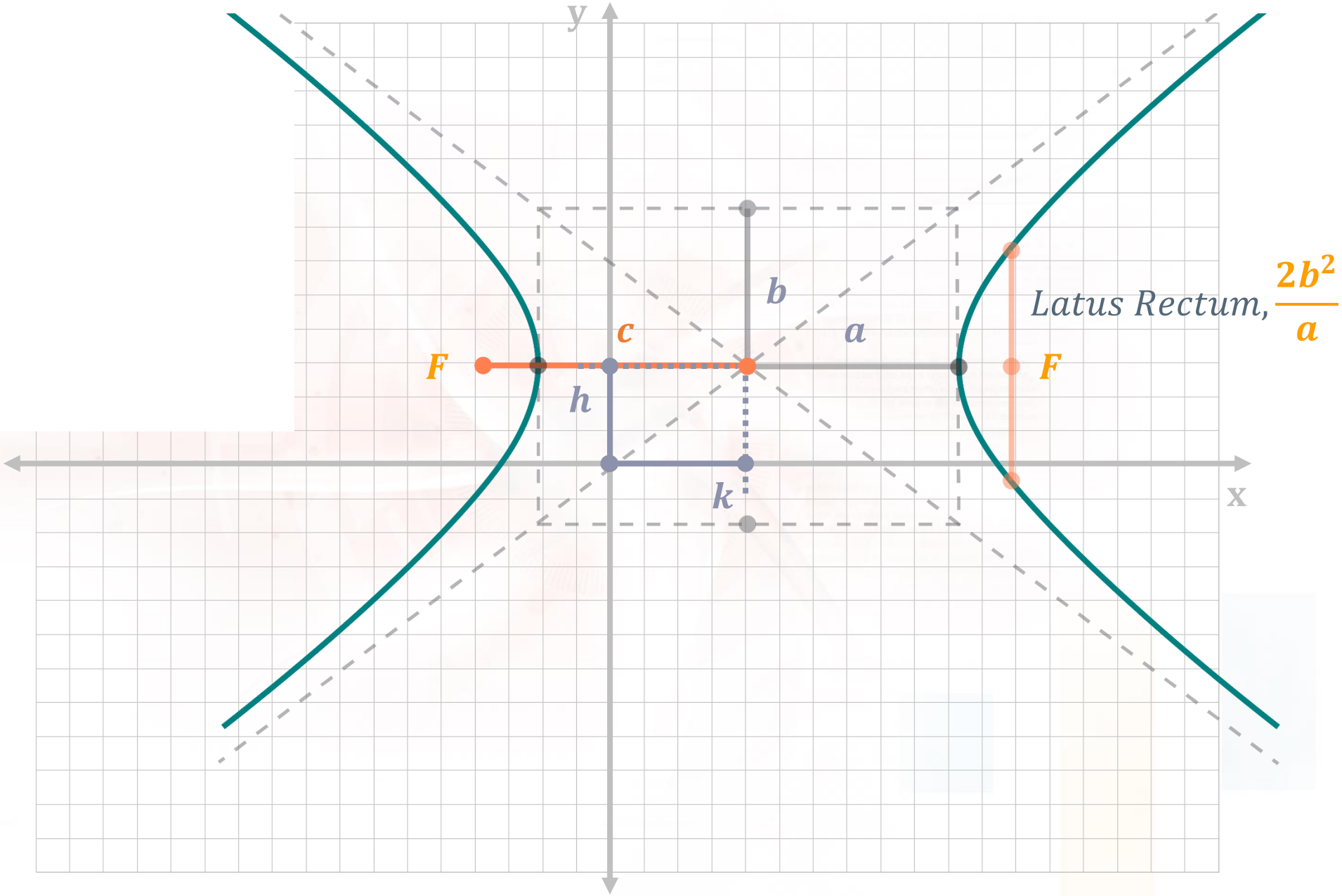


FOCI

foci,

$F(h \pm c, k)$

$c^2 = a^2 + b^2$



TRANSVERSE AXIS is along the $x - axis$

Standard Equation,

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(h, k)$

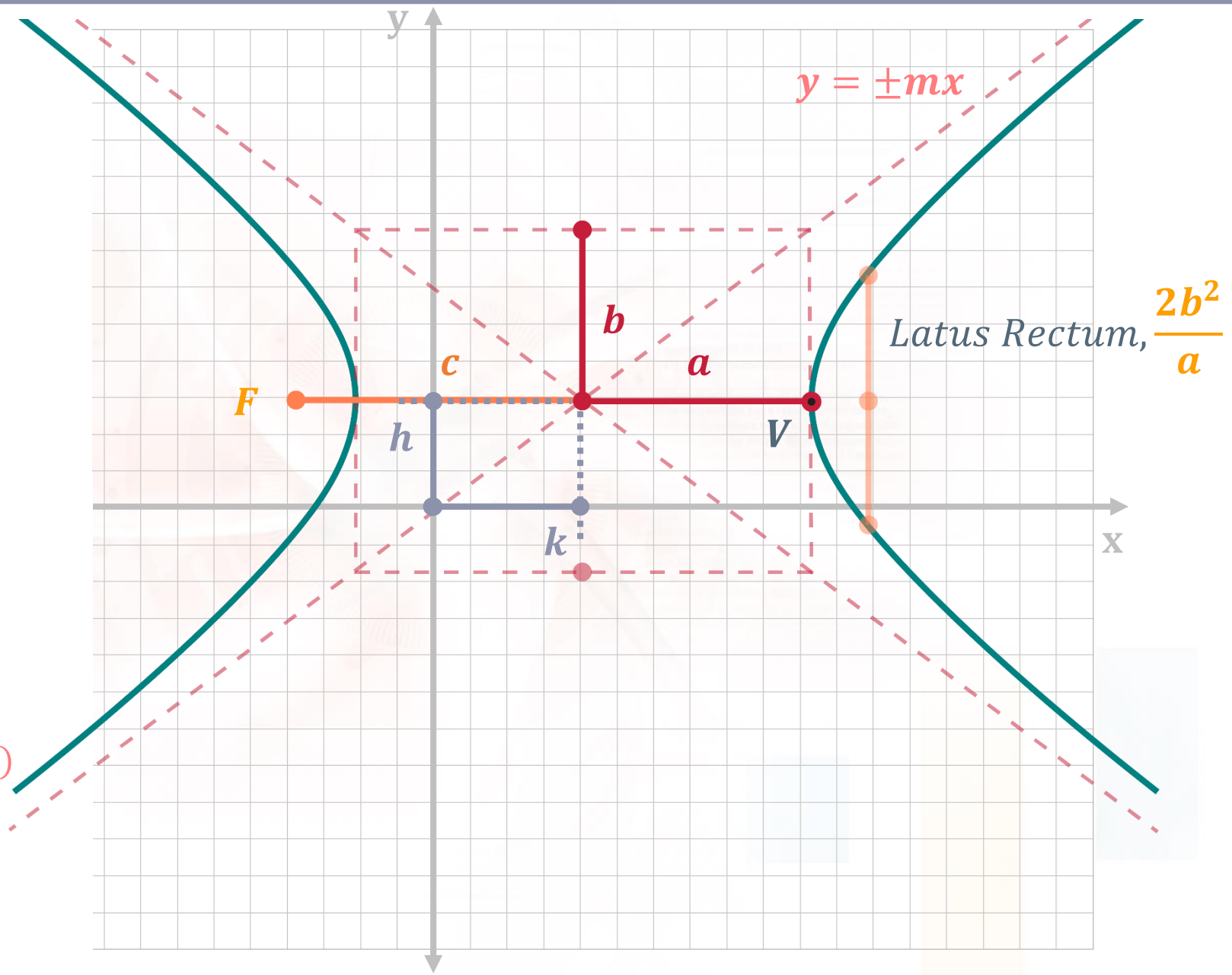
vertices, $V(h \pm a, k)$

foci, $F(h \pm c, k)$

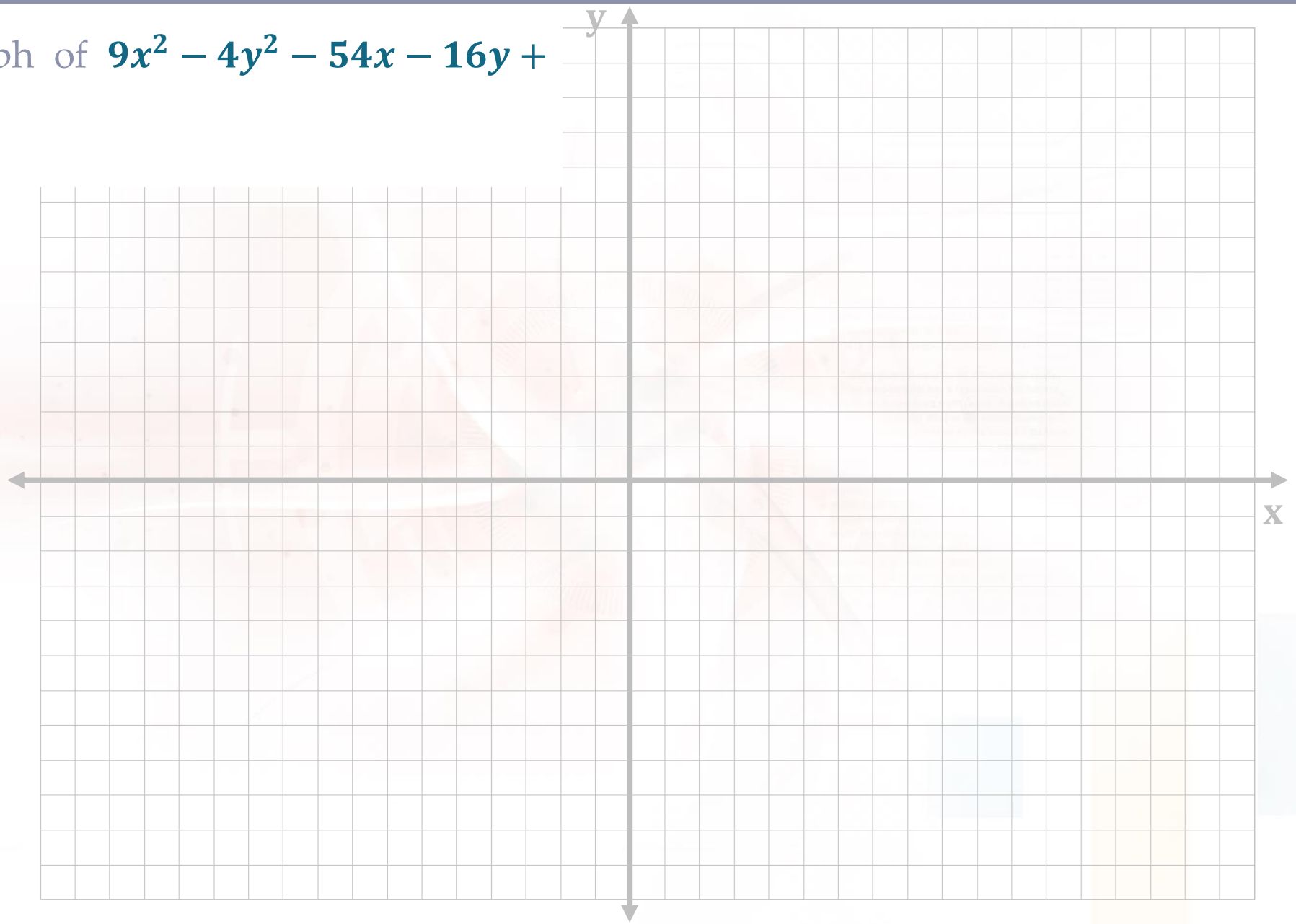
asymptotes, $(y - k) = \pm \frac{b}{a}(x - h)$

transverse axis, $2a$

conjugate axis, $2b$



Sketch the graph of $9x^2 - 4y^2 - 54x - 16y + 29 = 0$.



TRANSVERSE AXIS is along the $y - axis$

Standard Equation,

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(h, k)$

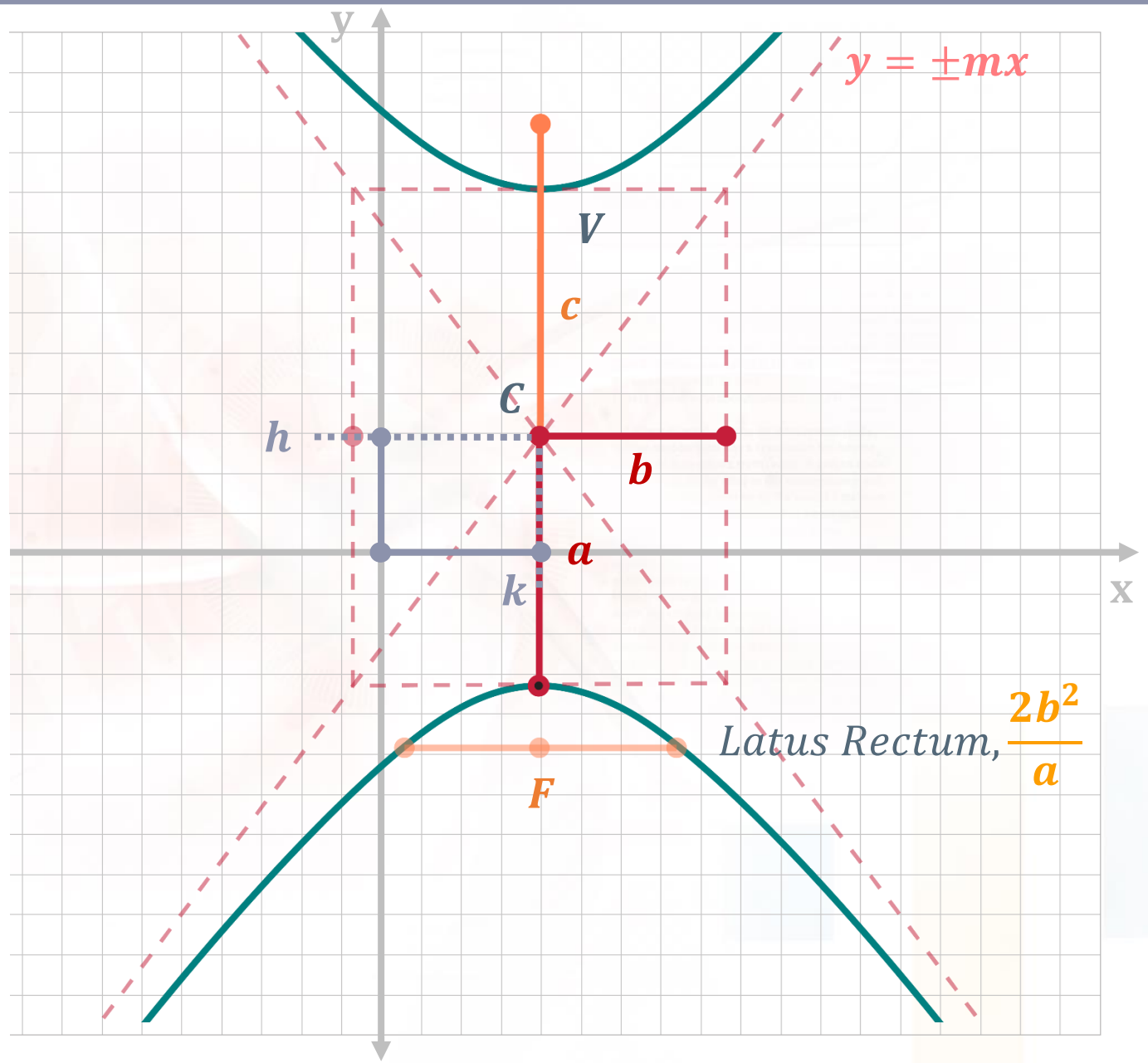
vertices, $V(h, k \pm a)$

foci, $F(h, k \pm c)$

asymptotes, $(y - k) = \pm \frac{a}{b}(x - h)$

transverse axis, $2a$

conjugate axis, $2b$



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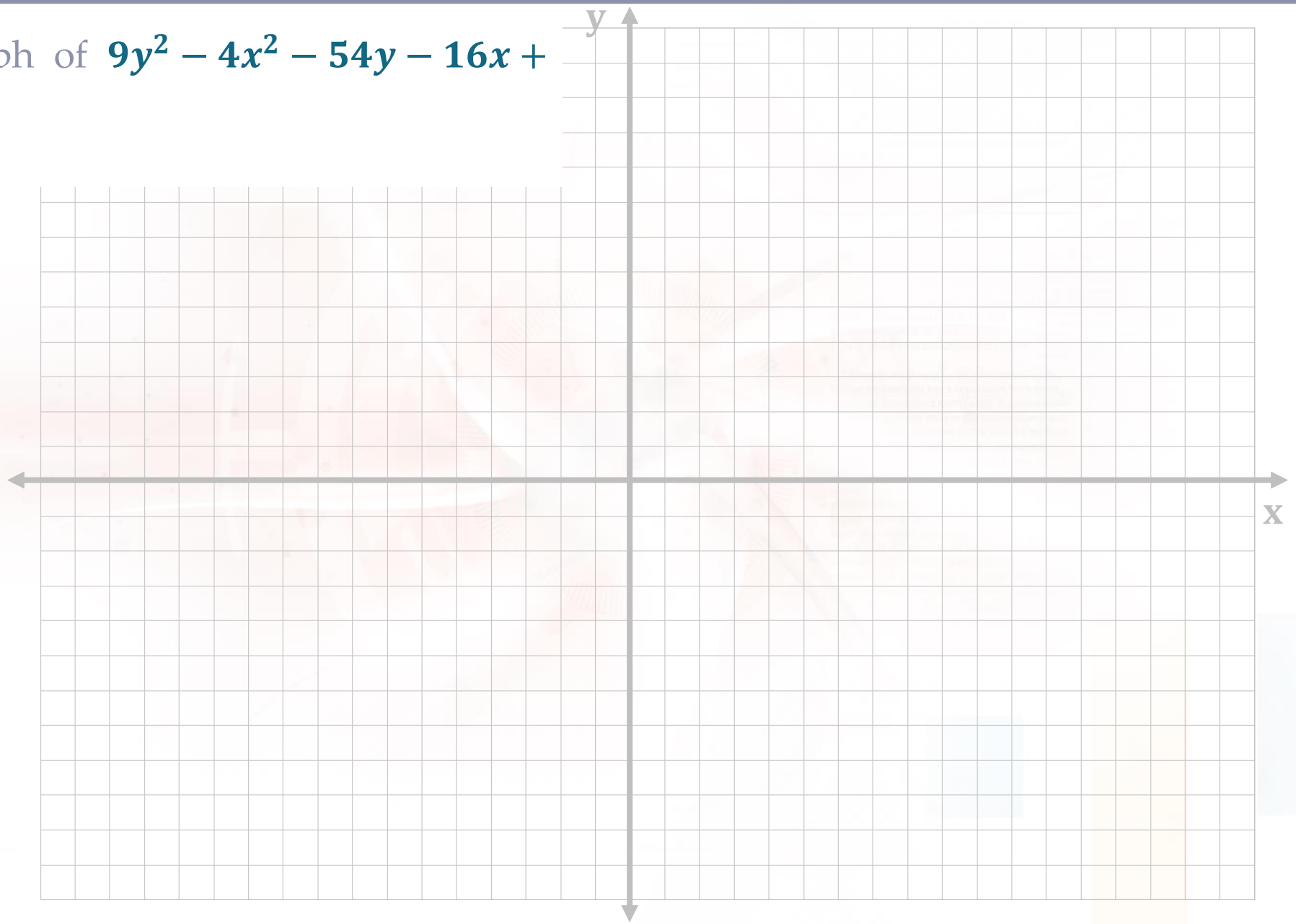
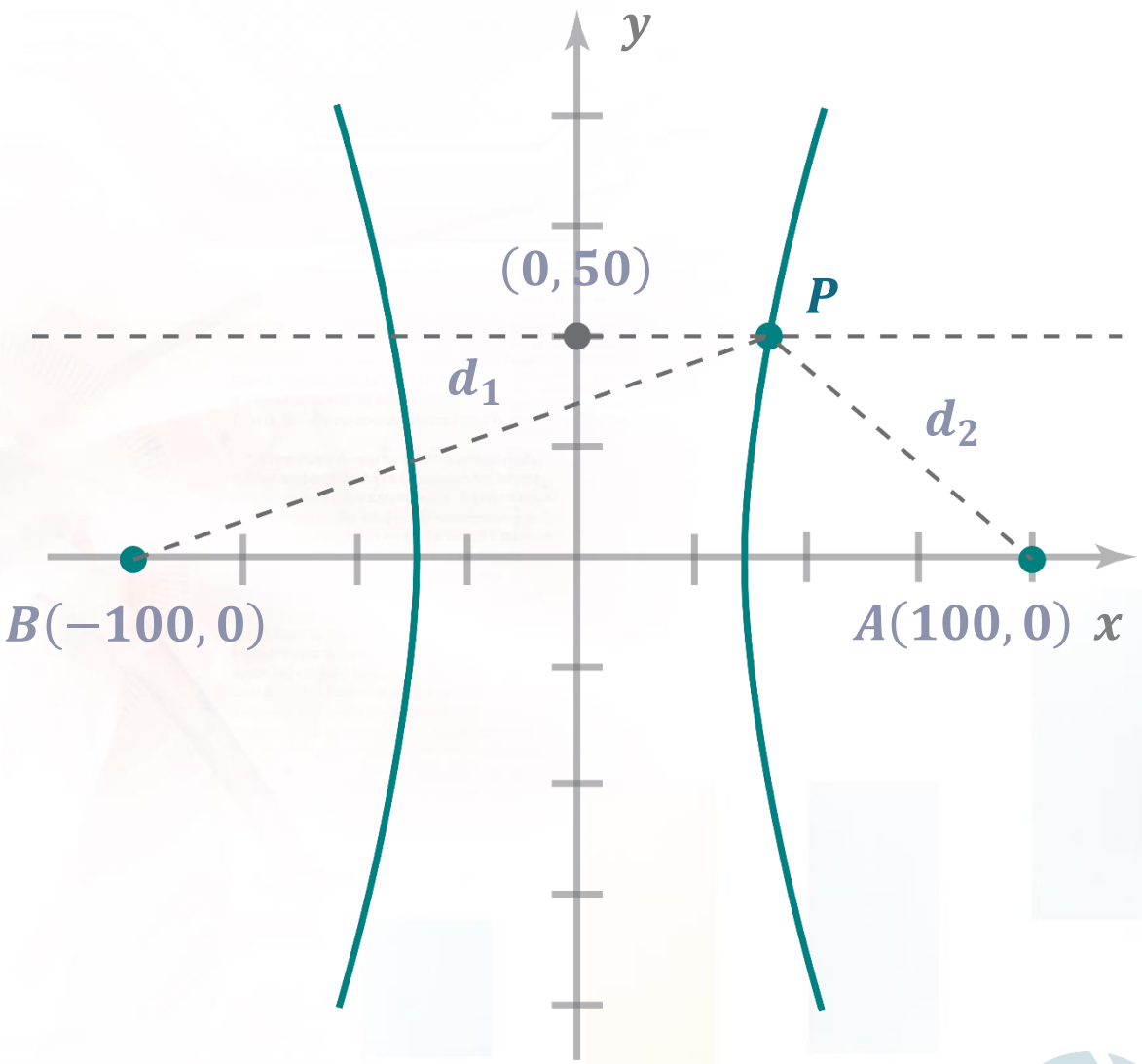


SEATWORK

1. Find the **standard equation** of the hyperbola

$$4y^2 - x^2 + 40y - 4x + 60 = 0$$

2. Find the **asymptotes** of the hyperbola

$$\frac{x^2}{49} - \frac{y^2}{16} = 1$$

3. Coast Guard **station A** is **200** miles directly east of another station B. A **ship** is sailing on a line **parallel** to and **50** miles north of the line through A and B. Radio signals are sent out from A and B at the rate of **980 ft/μsec** (microsecond). If, at 1:00 P.M., the signal **from B** reaches the ship **400 microseconds** after the signal from A, locate the **position** of the ship at that time.